Fiscal Year:	FY 2018	**Task Last Updated:**	FY 04/21/2021
PI Name:	Neelam, Srujana Ph.D.		
Project Title:	LINC Dependent Effect of Microgravity on Nuclear Shaping and Gene Expression in Breast Epithelial Cells		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NASA Postdoctoral Program
Start Date:	08/22/2016	**End Date:**	08/21/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Administration, USRA	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Extended to 8/21/2018 per H. Levine/KSC (original end date was 8/21/2017)--Ed., 10/29/18		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Zhang, Ye M.D., Ph.D. (MENTOR/ NASA Kennedy Space Center)		
Grant/Contract No.:	NASA Postdoctoral Program		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	NASA Postdoctoral Program Fellow Cells cultured in microgravity have altered cell morphology and gene expression. Cells reorganize their cytoskeleton when exposed to conditions of microgravity. What causes these effects is not entirely understood. Cells sense the external cues and these cues alter the cytoskeletal forces, which in turn are sensed by the nucleus due to the mechanical coupling between the nucleus and the cytoskeleton. In normal gravity, cell morphology is linked to both nuclear morphology and gene expression. LInker of Nucleoskeleton and Cytoskeleton (LINC) complex proteins are found to be involved in mechanosensing and maintaining normal nuclear shape and gene expression. Taken together, this study aims to understand how nuclear shape changes in microgravity in breast epithelial cells and what is its mechanism. Does LINC complex play a role in gravisensing? If so, then what is its role in nuclear shaping and how it regulates the sensitivity of genome-wide transcription to gravity?		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Exposure of cells to microgravity is known to lead to the reorganization of the cytoskeleton and the ECM (extracellular matrix). In addition, microgravity leads to morphological changes in both the cell and its nucleus, and to altered gene expression, all features related to LINC complex function. The LINC complex is also known to be a critical factor in the normal formation and development of tissues, such as the breast cell acinus. Simulated microgravity (SMG) disrupts features such as cell proliferation, cell adhesion, cell cycle progression, and cytokine production, themselves all linked to aberrant developmental regulation, such as cancer and which are primarily LINC dependent processes as cited above. Indeed, in addition to altering all these LINC-dependent features, SMG also suppresses tumor cell growth and development in e.g., breast epithelial cell cultures. However, despite such circumstantial evidence, the role of the LINC complex in cellular responses to altered gravitational environments has yet to be studied. Therefore, we have asked how the LINC complex affects changes in nuclear shape and differential gene expression in simulated microgravity.
Task Progress:	We first demonstrated that SMG alters nuclear morphology in MCF 10A cells. To understand if the change in nuclear shape in response to SMG depends on the LINC complex, we used a doxycycline inducible MCF 10A cell line where LINC complex function is disrupted. Cells expressing a similarly inducible GFP-KDEL were used as the corresponding paired controls. The induced cells were exposed to simulated microgravity on the 3d clinostat for 2 h and 20 h. The 1 x G controls (i.e., identically treated cells but not on the clinostat) were placed next to the clinostat in the same incubator. After treatment, the 3d nuclear morphology was monitored using confocal microscopy. Since the normal changes in nuclear shape triggered by SMG appear to require the LINC complex, we hypothesized that disruption of the LINC complex with SUN1L-KDEL expression might also affect patterns of gene expression. We used RNAseq analysis to characterize differential gene expression in GFP-KDEL and SUN1L-KDEL in SMG for 2 h and 20 h versus the corresponding 1 x G control. Larger number of genes were up- or down-regulated in SUN1L-KDEL cells in SMG compared to the GFP-KDEL cells which indicates a likely role of the LINC complex in regulating gene expression in SMG.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Articles in Peer-reviewed Journals	Neelam S, Richardson B, Barker R, Udave C, Gilroy S, Cameron MJ, Levine HG, Zhang Y. "Changes in nuclear shape and gene expression in response to simulated microgravity are LINC complex-dependent." Int J Mol Sci. 2020 Sep 15;21(18):E6762. https://doi.org/10.3390/ijms21186762 ; PMID: 32942630; PMCID: PMC7555797 , Sep-2020
Articles in Peer-reviewed Journals	Neelam S, Lee A, Lane MA, Udave C, Levine HG, Zhang Y. "Module to support real-time microscopic imaging of living organisms on ground-based microgravity analogs." Appl Sci. 2021 Apr;11(7):3122. https://doi.org/10.3390/app11073122 , Apr-2021
Articles in Peer-reviewed Journals	Zhang Y, Moreno-Villanueva M, Krieger S, Ramesh GT, Neelam S, Wu H. "Transcriptomics, NF-kappaB pathway, and their potential spaceflight-related health consequences." Int J Mol Sci. 2017 May 31;18(6):E1166. Review. https://doi.org/10.3390/ijms18061166 ; PMID: 28561779; PMCID: PMC5485990org , May-2017
Articles in Peer-reviewed Journals	Thiel CS, Tauber S, Seebacher C, Schropp M, Uhl R, Lauber B, Polzer J, Neelam S, Zhang Y, Ullrich O. "Real-time 3D high-resolution microscopy of human cells on the International Space Station." Int J Mol Sci. 2019 Apr 25;20(8):E2033. https://doi.org/10.3390/ijms20082033 ; PMID: 31027161; PMCID: PMC6514950 , Apr-2019
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Articles in Peer-reviewed Journals	Dixit AR, Meyers AD, Richardson B, Richards JT, Richards SE, Neelam S, Levine HG, Cameron MJ, Zhang Y. "Simulated galactic cosmic ray exposure activates dose-dependent DNA repair response and down regulates glucosinolate pathways in Arabidopsis seedlings." Front Plant Sci. 2023 Dec14;14:1284529. https://doi.org/10.3389/fpls.2023.1284529 , Dec-2023
Books/Book Chapters	Zhang Y, Richards JT, Hellein JL, Johnson CM, Woodall J, Sorenson T, Neelam S, Ruby AMJ, Levine HG. "NASA's ground-based microgravity simulation facility." in "Plant Gravitropism. Methods in Molecular Biology" Ed. E.B. Blancaflor. New York, NY: Humana, 2022. vol. 2368: pp. 281-99. https://doi.org/10.1007/978-1-0716-1677-2_18 ; PubMed PMID: 34647262, Jan-2022